

Biology: Cells and Energy

The cell is the basic structural and functional unit of life. The cell membrane controls movement of substances into and out of the cell. The cytoplasm is where many chemical reactions take place.

The nucleus stores most of a eukaryotic cell's genetic material and helps regulate cell activities. Mitochondria release usable energy from food molecules during cellular respiration.

Photosynthesis occurs mainly in chloroplasts. Plants use light energy to convert carbon dioxide and water into glucose and oxygen. The balanced summary equation is carbon dioxide plus water, in the presence of light, produces glucose and oxygen.

Cellular respiration transfers energy from glucose into ATP, which cells use for work. Respiration occurs in plants and animals; photosynthesis is limited to organisms that have photosynthetic pigments.